BUSINESS ANALYTICS CASE STUDY

Understanding Customer Churn at a Telecommunications Company

Telco Customer Churn Dataset · 7,043 Customers · 21 Variables

AnalystLab Africa Data Analytics Internship · Batch B · Week 5

Problem Statement

Why are customers leaving, and which factors matter most?

1

The Challenge

The company is losing a significant share of customers to churn, but retention resources are limited and must be spent where they matter most.

2

The Questions

1. Why are customers leaving the company?
2. Which factors contribute most to churn, and which segments should be prioritized for retention?

Dataset Overview

7,043
Customer Records

21
Variables

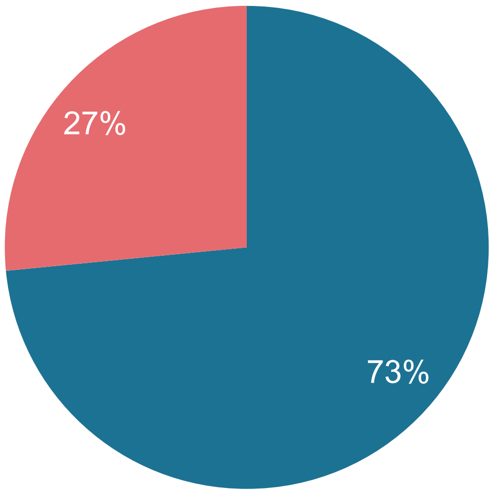
0
Duplicate Rows

26.54%
Overall Churn Rate

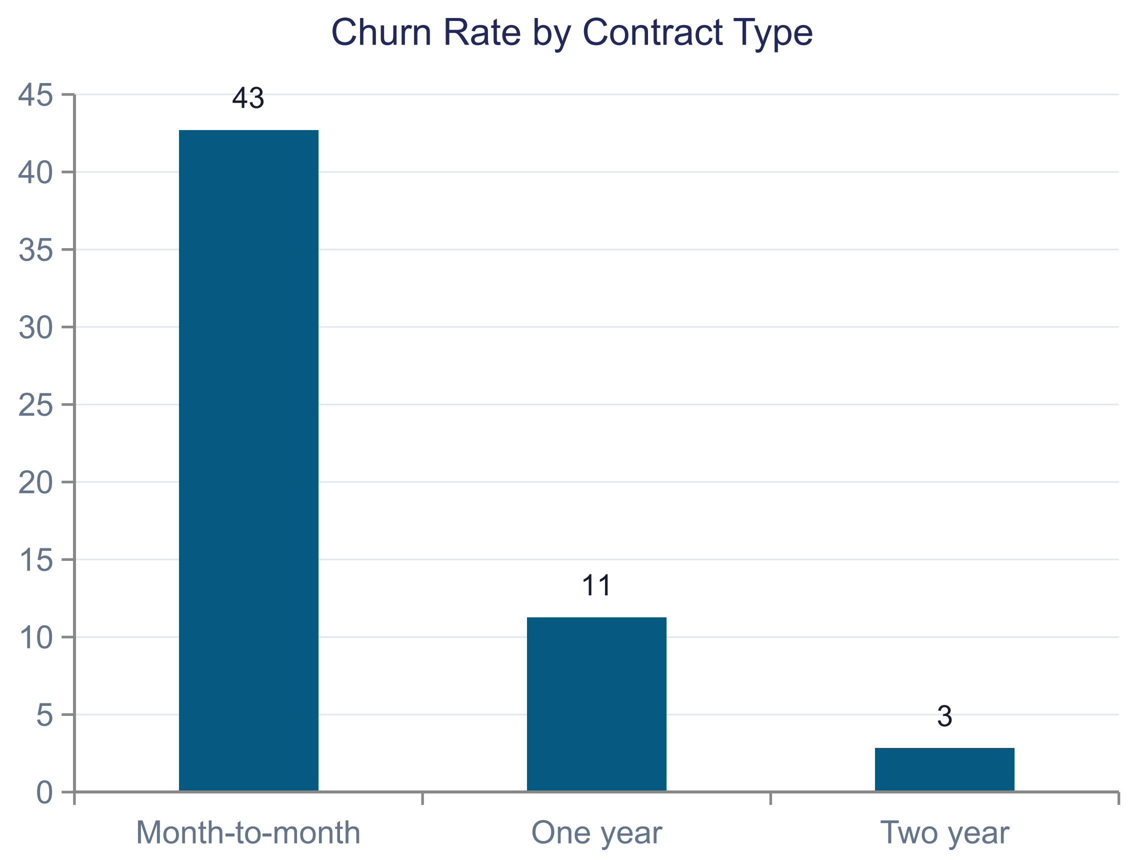
Variable Categories

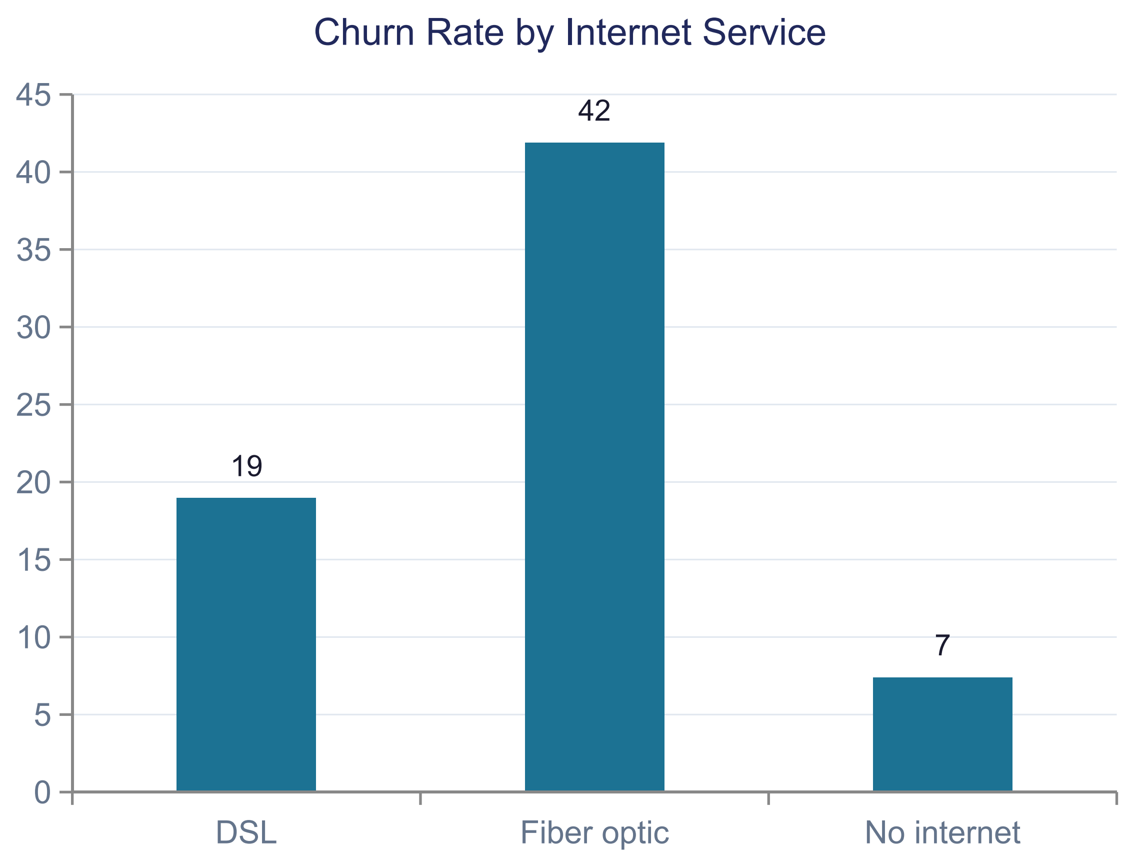
- Demographics: gender, senior citizen, partner, dependents
- Account: tenure, contract type, payment method, billing
- Services: internet, security, tech support, streaming
- Charges: monthly charges, total charges

Overall Churn Split (%)



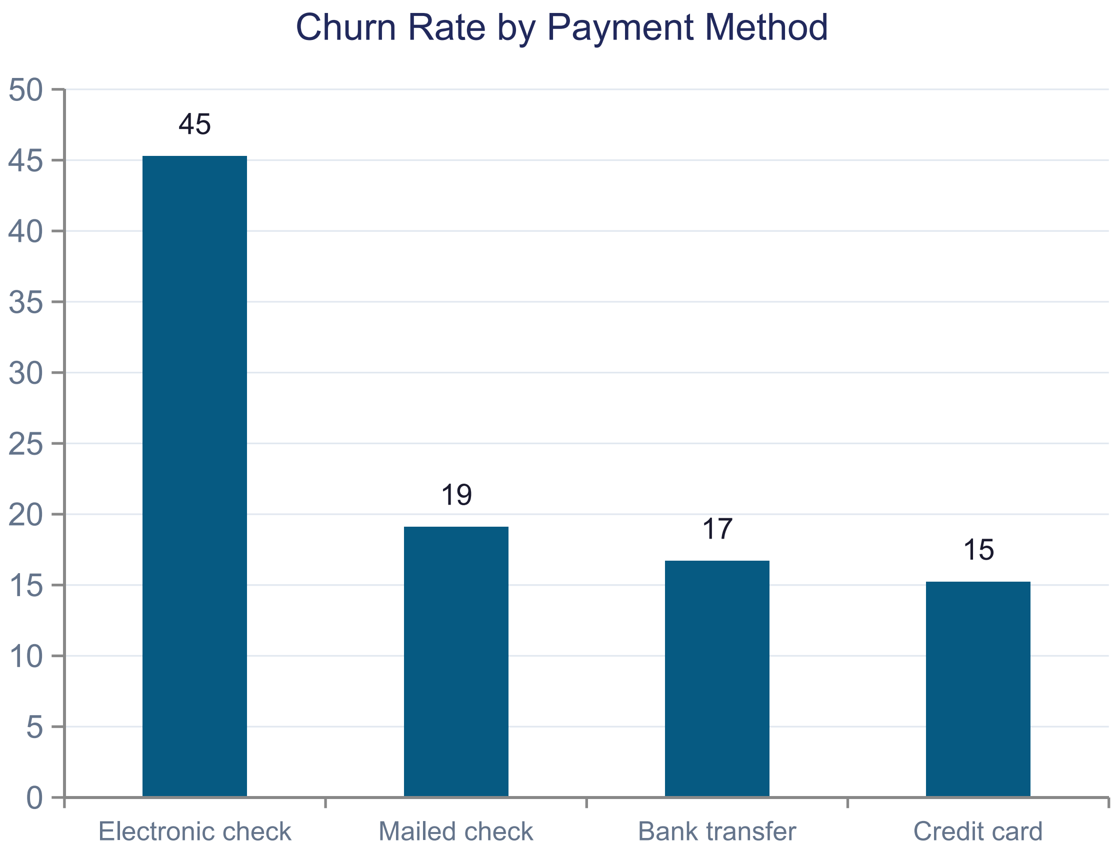
Churn Rate by Contract & Internet Service

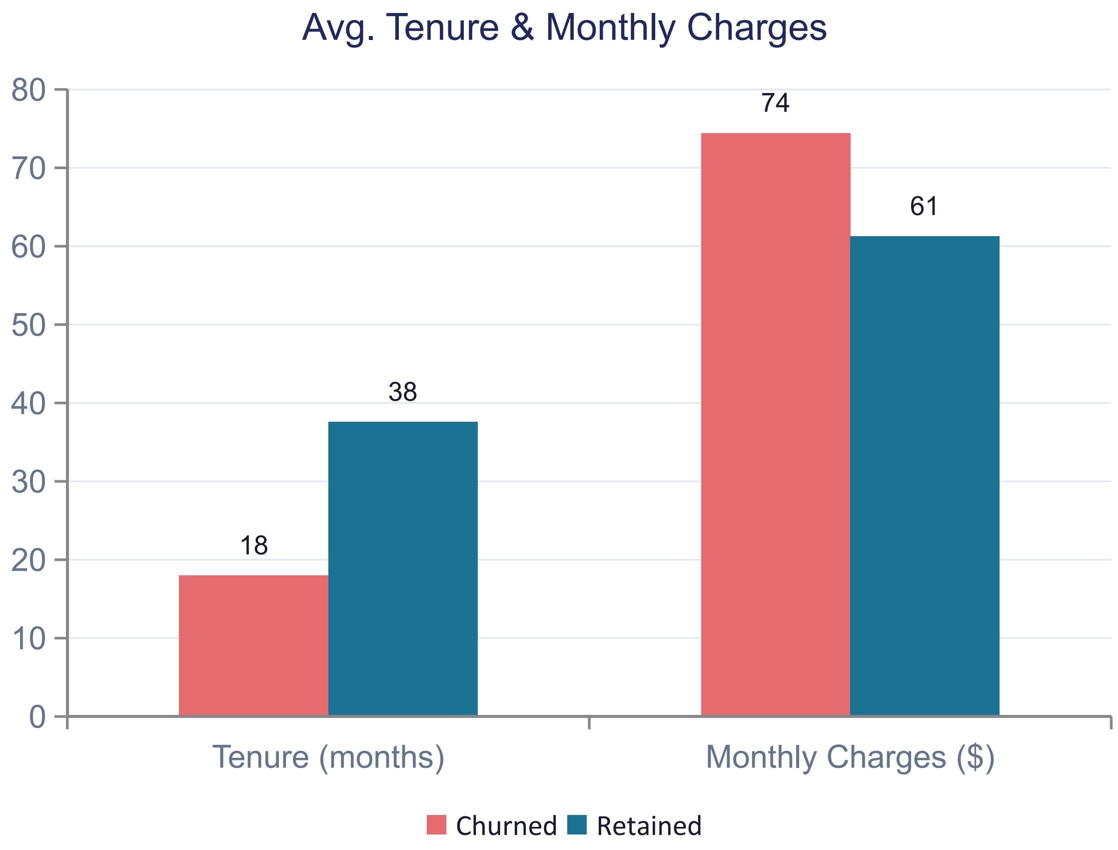




Month-to-month customers churn ~15x more than two-year customers. Fiber optic customers churn nearly 3x more than DSL customers.

Payment Method & Customer Tenure





Electronic check payers churn ~3x more than autopay customers. Churned customers leave at half the tenure of retained ones, despite paying more per month.

Major Findings

01

Contract type is the strongest driver

Churn falls from 42.7% (month-to-month) to 2.8% (two-year contracts).

02

Premium service, higher churn

Fiber optic customers churn nearly 3x more than DSL, despite paying more.

03

Payment friction matters

Electronic check users churn ~3x more than customers on autopay.

04

New customers are most at risk

Churned customers average 18 months tenure vs. 38 months for retained.

05

Add-ons build loyalty

Customers without online security or tech support churn 2.5-3x more.

06

Demographics shape risk

Seniors, and customers without a partner or dependents, churn more.

Recommendations

1. Incentivize longer contracts, prioritizing month-to-month customers in their first 12 months.
2. Investigate the fiber optic experience — pricing, reliability, and competitive positioning.
3. Migrate electronic check payers toward automatic payment methods.
4. Bundle and proactively promote online security and tech support add-ons.
5. Build an early-tenure retention program focused on a customer's first year.
6. Tailor retention outreach for senior citizens and single customers without dependents.

Conclusion

Churn at this company is not random. It is concentrated among new, month-to-month, fiber optic customers who pay by electronic check and lack add-on services. Contract type, tenure, service type, payment method, and add-ons are the clearest, most actionable levers identified.

Acting on these findings offers a concrete, data-driven path to reducing the 26.54% churn rate and protecting recurring revenue.

Thank you